Architecture: v1 Synthetic CRM CSVs + Recommender

Overview
This v1 system generates a realistic synthetic CRM dataset, builds modeling-ready feature views, tra

Goals
• Generate realistic synthetic CRM data as versioned CSVs.
• Model a Salesforce-like and Outreach-like GTM environment.
• Commit generated datasets into gtm-experimentation.
• Support future migration to Postgres, SQLite, DuckDB, Redis, MCP orchestration, and a web UI.

Non-Goals
• Production authentication
• Multi-tenant isolation
• PII handling
• Real LLM orchestration
• Full web frontend

Core Workflow
1. Generate synthetic CRM data
2. Save CSVs
3. Build feature views
4. Train regression model
5. Generate recommendations
6. Explain recommendations

System Components
• Chat / CLI UI
• Orchestrator
• Tool Router
• CSV Data Generator
• CSV Store
• Feature Builder
• Model Trainer
• Recommender
• Recommendation Explainer

Data Model

CRM Tables
accounts.csv
- account_id
- account_name
- industry
- employee_count
- region
- tools_used
- innovation_adoption
- risk_tolerance
- budget_band
- competitor_set

contacts.csv
- contact_id
- account_id
- first_name
- last_name
- persona
- seniority
- department
- email
- phone

sellers.csv
- seller_id
- seller_name
- role
- territory
- capacity
- style_traits

opportunities.csv
- opportunity_id
- account_id
- primary_contact_id
- owner_seller_id
- stage

- amount
- created_at
- target_close_date
- outcome

Activity Tables
activities.csv
call_dispositions.csv
calls.csv

Knowledge Graph Tables
kg_entities.csv
entity_mentions.csv

Market Intelligence
market_signals.csv

Determinism & Reproducibility
The generator uses a seed and deterministic configuration so identical inputs produce identical outpu

Recommendation Engine
Target:
• Opportunity stage advancement
• Opportunity win probability

Features:
• Engagement metrics
• Contact coverage
• Activity volume
• Sentiment signals
• Objection counts
• Competitor mentions
• Market signals
• Firmographic and psychographic attributes

Outputs:
• Top-N recommendations
• Predicted score
• Key contributing factors
• Human-readable explanation

Repository Structure

```
gtm-experimentation/
  architecture.md
  configs/
  data/
    synth_csv/
    feature_views/
  models/
  src/
    orchestration/
    data_generation/
    features/
    modeling/
    cli/
  tests/
```

Commit Policy

Commit:
• architecture.md
• configs
• synthetic datasets
• training code
• recommendation code
• tests

Do Not Commit:
• .env
• secrets
• API keys
• real customer data

Future Extensions
• Postgres / SQLite / DuckDB
• Redis caching

- MCP orchestration
- LLM reasoning
- Web UI
- Evaluation dashboards
- Scenario simulation

Success Criteria
- Generate coherent synthetic CRM data
- Build feature views
- Train regression model
- Produce recommendations
- Explain recommendations
- Run end-to-end via CLI/chat